Strengthening the Foundations of Law as Extended Phenotype: Coalitional Intelligence and the Evolutionary Origins of Legal Institutions

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Abstract

Evolutionary legal theory faces a fundamental explanatory gap: why natural selection produced organisms capable of creating, transmitting, and enforcing abstract normative rules. To my knowledge, this paper provides the first integration of the Coalitional Intelligence Hypothesis (CIH) by Egeland, Kennair, and Kleppestø (2026) with Extended Phenotype Theory (EPT) as applied to legal systems, generating a four-stage evolutionary genealogy that neither framework produces independently.

The CIH posits that human intelligence evolved as an honest signal of coalitional value, facilitating reciprocal exchanges of prestige for cognitive-computational services (conflict resolution, collective coordination, normative articulation). EPT argues that law operates as the extended phenotype of competing normative memes. The integration proposes that formal legal institutions are memetically codified elaborations of the coalitional value system: proto-jurists provisioning coordination services received prestige in small-scale societies; modern law institutionalizes that exchange through deontic language and specialized structures.

The integrated framework generates three falsifiable predictions with specified measurement protocols: (1) legal systems exhibit institutional signatures of coalitional value exchange; (2) professional prestige correlates with cognitive-computational service quality beyond formal credentials; (3) legal transplant failures correlate with coalitional value system mismatches. Boundary conditions are specified determining when CIH versus EPT have greater explanatory leverage, and limitations regarding indirect archaeological evidence and dependence on untested CIH predictions are explicitly acknowledged.

The integration provides evolutionary microfoundations for institutional legal theory, with testable implications for institutional design, legal reform, and comparative analysis of legal systems.

Keywords: *Extended Phenotype Theory, Coalitional Intelligence Hypothesis, evolutionary jurisprudence, legal origins, coalitional value, prestige, cognitive-computational services, reciprocal altruism, cultural evolution, memetic competition*

JEL Classification: K00, D03, Z13, C73, B52

1. Introduction: The Missing Mechanism

The question of why human societies developed formal legal institutions remains one of the most consequential unsolved problems in evolutionary social science. Legal theory has traditionally treated this question as either self-evident (positivism: law exists because sovereigns command) or transcendent (natural law: law exists because moral truths require institutional expression). Neither answer satisfies the basic Darwinian demand for a causal account of origins: what selection pressures produced organisms capable of creating, transmitting, and enforcing abstract normative rules?

In prior work, I have approached this question from two complementary directions. In "Before the Command Was Spoken" (Lerer, 2025a), I argued that law is not a human invention *ex nihilo* but the extended phenotype of pre-linguistic normative patterns that evolved to stabilize cooperation in early human groups. Drawing on primatological evidence (absence of third-party punishment in chimpanzees, coalition dynamics in bonobos), archaeological data (the Paleolithic egalitarian signature in burial practices), linguistic universals (deontic modality as ontogenetically primary), and ethnographic documentation (the !Kung San "insulting the meat" ritual), I argued that proto-norms predated prescriptive language and exerted selective pressure for its emergence. In "The Altruism Paradox in Law" (Lerer, 2025b), I argued that commercial law emerges through three mechanisms paralleling biological evolution: kin selection (generating trust networks in early merchant communities), reciprocal altruism (creating self-enforcing contracts and trade customs), and costly signaling (establishing reputational systems enabling anonymous exchange). The *lex mercatoria*, Jewish diamond merchant networks, Islamic *hawala* systems, and contemporary international arbitration all exhibit the same evolutionary pattern: cooperative legal orders emerge from iterated interactions among self-interested actors who develop legal norms as extended phenotypes of their survival strategies.

These two contributions established the *what* and the *how* of legal evolution. Proto-norms preceded language and created the substrate upon which formal law was built (*what*). Reciprocal altruism and costly signaling provide the mechanisms through which cooperation scales into institutional form (*how*). But a specific explanatory gap remained: *why* did human evolution produce individuals capable of provisioning the cognitive services that constituted the raw material of law? Why did natural selection favor the particular constellation of cognitive capacities (conflict resolution, collective action coordination, norm articulation, strategic planning) that proto-legal specialists deployed?

A recent paper by Egeland, Kennair, and Kleppestø (2026) in *Evolution and Human Behavior* offers a plausible answer. Their Coalitional Intelligence Hypothesis (CIH) posits that human intelligence evolved as an honest, status-enhancing signal of coalitional value, facilitating adaptive exchanges of prestige and cognitive-computational services within human groups. Building on Alexander's (1990) Ecological Dominance-Social Competition model, Dunbar's Social Brain Hypothesis, and Winegard et al.'s (2020) Coalitional Value Theory, the CIH argues that intelligent individuals provision services (conflict resolution, access to public goods, solutions to collective action problems) that increase the marginal fitness value of their coalitions. In return, they receive deference and status, establishing reciprocal relationships of mutually beneficial exchange.

The CIH addresses the explanatory gap in EPT's account of legal origins. The cognitive-computational services that Egeland et al. describe, documented ethnographically among forager-horticulturalists (Garfield and Hagen, 2024; Gurven et al., 2000; Neel, 1980), closely parallel the proto-legal services that I documented in "Before the Command Was Spoken": mediation of disputes, coordination of collective activities, distribution of resources, and articulation of norms governing cooperation. The CIH proposes why natural selection produced individuals capable of these services; EPT proposes how those services were formalized, memetically codified, and institutionalized into what we recognize as law.

This paper proceeds as follows. Section 2 reconstructs the CIH and identifies four specific contact points with EPT. Section 3 develops the integration, showing how the coalitional value system provides microfoundations for the extended phenotype framework. Section 4 addresses the institutional scaling problem: how services provisioned by intelligent individuals in small groups became formalized legal systems governing millions. Section 5 derives three novel predictions from the integrated framework and operationalizes them with falsification criteria. Section 6 discusses limitations and future directions.

2. The Coalitional Intelligence Hypothesis and Its Contact Points with Extended Phenotype Theory

Egeland et al. (2026) develop the CIH in response to a recognized gap in the literature on human cognitive evolution. While multiple converging lines of evidence establish that social selection pressures drove the evolution of human intelligence (Alexander, 1989; Crespi et al., 2022; Dunbar, 1998, 2024; Flinn et al., 2005), existing hypotheses do not specify with sufficient precision how greater cognitive ability proved beneficial in social competition. The Machiavellian Intelligence Hypothesis (Byrne and Whiten, 1988) focuses on manipulation and deception; the Vygotskyan Intelligence Hypothesis (Moll and Tomasello, 2007) focuses on cooperation. Both treat competition and cooperation as mutually exclusive. The CIH resolves this by showing that competition and cooperation are simultaneously operative in the coalitional value system.

The core mechanism proposed by Egeland et al. can be summarized as follows. Human ancestral groups solved the coordination problem of reconciling individual fitness-relevant advantages with social cooperation by instituting a system in which status (specifically prestige, not dominance) is conferred upon individuals whose traits or behaviors provide the coalition with benefits (Anderson et al., 2015; Henrich and Gil-White, 2001; Price and Van Vugt, 2014a, 2014b; Winegard et al., 2020). Intelligent individuals provision cognitive-computational services that improve group coordination, and in return they receive deference from others. This would establish reciprocal relationships where both parties benefit: the intelligent individual gains fitness through increased access to mates, cooperation partners, and resources; the group gains fitness through improved collective action. The behavioral manifestations of intelligence function as honest signals of coalitional value, because only genuinely competent individuals can sustain the provision of services under the accountability demands of repeated interaction within stable coalitions.

Four specific contact points between the CIH and EPT merit detailed analysis.

2.1. Cognitive-Computational Services as Proto-Legal Functions

Egeland et al. cite extensive cross-cultural evidence that intelligence ranks among the most valued criteria for status allocation. Data across 14 countries show intelligence as the second most important criterion (of 240 items) for assigning high status (Buss et al., 2020). Among forager-horticulturalists, traits crucial for coalitional cohesion and success, such as conflict resolution skills, technical abilities, food sharing, trading skills, and healthcare provisioning, constitute central criteria for status allocation and are predicted by peer-rated intelligence (Garfield and Hagen, 2024; Gurven et al., 2000).

These ethnographic findings describe, in the vocabulary of evolutionary psychology, exactly what I identified in "Before the Command Was Spoken" as proto-legal services. The *headman* documented by Neel (1980) among the Yanomami, the mediator among the Tsimane studied by von Rueden et al. (2019), the conflict resolver among Ethiopian hunter-gatherers documented by Garfield and Hagen (2024): all are individuals who receive prestige in exchange for provisioning cognitive services that solve coordination problems. They can be understood as proto-jurists. Their function, stripped of modern institutional scaffolding, closely parallels what courts, legislators, and legal advisors do in contemporary societies: they process information about disputes, norms, and collective goals, and they output coordination solutions that others defer to.

The connection to EPT follows naturally. If legal institutions are extended phenotypes of normative memes (Lerer, 2025a, 2026a), and if those memes originally encoded solutions to coordination problems provisioned by intelligent individuals in exchange for prestige (Egeland et al., 2026), then the evolutionary origin of law can be understood as the formalization of the coalitional value exchange. Law, on this account, is the institutionalized, memetically transmitted, and phenotypically extended version of the cognitive-computational services that the CIH describes.

2.2. The Coalitional Value Gauge and Heteronomous Bayesian Updating

Egeland et al.'s second prediction states that humans can gauge the intelligence levels of others with significant accuracy, even from minimal social interaction. This capacity is a prerequisite for the coalitional value system: status can be conferred based on coalitional value only if that value can be assessed. Multiple studies confirm this capacity (Borkenau et al., 2004; Driebe et al., 2021; Zebrowitz et al., 2002).

In the EPT framework, this gauging capacity has an institutional analogue that I have formalized as Heteronomous Bayesian Updating (HBU). HBU describes how legal professionals learn which norms are "correct" not by observing outcomes (whether a legal rule produces good social results) but by observing reactions of authority figures (whether a judge, professor, or senior colleague approves or disapproves of a given interpretation). These reactions are, in the terminology of Egeland et al., evaluations of coalitional value transmitted vertically through institutional hierarchies.

The parallel is suggestive. In small-scale societies, a young individual calibrates their assessment of who provides valuable cognitive services by observing community reactions to different individuals' advice. In a law school, a first-year student calibrates their assessment of which legal doctrines are "correct" by observing the reactions of professors to different answers. Both processes are instances of the coalitional value gauge operating through observed social reactions rather than through direct outcome evaluation. HBU may represent the institutional elaboration of the same evolved mechanism that Egeland et al. describe at the individual level.

The critical difference, however, is that in formalized legal systems, HBU can decouple from genuine cognitive capacity. A lawyer may receive deference not because they resolve coordination problems better than alternatives, but because they replicate the dominant memeplex with high fidelity. This decoupling is a specific form of the cheater problem that Egeland et al. acknowledge as a threat to the coalitional value system, but which acquires distinctive institutional characteristics in legal contexts. I return to this point in Section 4.

2.3. Runaway Social Selection and Constitutional Lock-in

Egeland et al. develop Alexander's (1990) model of runaway social selection: when social interaction becomes the principal source of selection (ecological dominance), traits facilitating success in social competition are amplified through within-species arms races. The selection pressures favoring intelligence and the selection pressures favoring deference toward intelligent individuals become mutually reinforcing, producing the accelerating dynamics that Fisher (1930) identified for sexually dimorphic phenotypes.

This mechanism has a precise institutional analogue in what I measure with the Constitutional Lock-in Index (CLI). Argentina maintained a national income tax for 93 years without constitutional authorization, a globally unique persistence that I analyzed as an extended phenotype of federalist identity memes (Lerer, 2025c). In systems of high constitutional rigidity (Argentina, CLI approximately 0.89), institutional competition becomes endogenous. Actors compete not against the external environment but against each other for control of veto mechanisms, blocking coalitions, and institutional rents. Legal professionals benefit from doctrinal stability (which protects the value of their expertise). Judges benefit from review powers (which enhance institutional prestige). Organized interests benefit from veto points (which enable rent extraction). Each actor's strategy reinforces the others', producing the institutional equivalent of runaway selection: the system becomes increasingly rigid not because rigidity serves any external adaptive function, but because the internal dynamics of status competition among institutional actors make it so.

The connection to EPT's concept of Parasitic Spontaneous Order (PSO) is suggestive. When the coalitional value system described by Egeland et al. becomes institutionally embedded, the prestige economy can become parasitic: institutional actors accumulate status by defending existing configurations (active inference, in Friston's terminology) rather than by provisioning genuinely useful cognitive services (perceptual inference). The extended phenotypes of the dominant memeplex, tribunals, law school curricula, bar associations, professional journals, then function not to maximize coalitional fitness but to minimize the cost of belief revision for incumbent actors. The system generates runaway institutional selection that serves the replication interests of established memes rather than the coordination needs of the group.

2.4. Encoding Coalitional Value: Citations as Reputational Signals

Egeland et al.'s third prediction, that information about intelligence levels is better encoded than evolutionarily irrelevant information, has not yet been tested experimentally. The authors note that human memory is often adaptive, with evolutionarily relevant information encoded more strongly than other kinds (Nairne et al., 2008; Xu et al., 2024), and that social information enjoys encoding advantages over ecological information (Redhead and Dunbar, 2013).

In the EPT framework, this prediction translates into a specific institutional phenomenon: the persistence of reputational information about legal actors across generations. The citation of authority in legal discourse (invoking the doctrines of Borda, Llambias, or Morello in Argentine civil law; Cardozo, Holmes, or Posner in American jurisprudence) is a system of encoding coalitional value that crystallizes individual cognitive contributions into transmissible textual form. A legal citation is, functionally, a reputational signal: it communicates that the cited author provided cognitive-computational services of sufficient quality to merit deference from subsequent decision-makers.

This connects to the Transmission Capacity Index (TCI) that I have developed as part of the EPT measurement framework. Legal systems that encode information about cognitive coalitional value with higher fidelity (through robust stare decisis, identified doctrinal authorship, accessible jurisprudential databases) should exhibit greater memetic transmission capacity than systems with opaque mechanisms of prestige allocation. The prediction is testable: common law systems, with their tradition of named judicial opinions and explicit precedent-following, should score higher on TCI than civil law systems where doctrinal contributions are often anonymized within codified texts.

3. The Integration: From Coalitional Value to Institutional Law

The integration of the CIH with EPT produces a four-stage genealogical account of legal origins that is more complete than either framework achieves alone.

Stage 1: Ecological Dominance and the Coalitional Value System. As human ancestors became ecologically dominant (Alexander, 1990; Flinn et al., 2005), intraspecific social interaction became the principal source of selection. Groups that solved the coordination problem, reconciling individual fitness advantages with collective cooperation, outcompeted those that did not. The solution was a system of reciprocal exchange where prestige was conferred on individuals whose traits or behaviors provided coalitional benefits (Egeland et al., 2026; Winegard et al., 2020). Among the most valued services were cognitive-computational contributions: conflict resolution, strategic planning, collective action coordination, and norm articulation. This is the CIH's contribution.

Stage 2: Proto-Norms as Pre-Linguistic Phenotypes. The cognitive-computational services provisioned by intelligent individuals were not isolated performances but patterned behavioral regularities sustained by third-party sanctions. As I documented in "Before the Command Was Spoken" (Lerer, 2025a), these proto-norms operated through embodied practice (the !Kung "insulting the meat" ritual), partner selection (chimpanzee collaboration partner choice; Melis et al., 2006), and coalition dynamics (bonobo female reverse dominance hierarchies; Surbeck et al., 2025). They functioned without prescriptive language, relying on what Dennett (2017) calls "competence without comprehension." The normative patterns were extended phenotypes of behavioral memes, environmental modifications that enhanced the replication of cooperative strategies. This stage fills a gap in the CIH: Egeland et al. do not specify how the cognitive-computational services described in their model were encoded and transmitted before language. A plausible answer is proto-normative phenotypes.

Stage 3: Deontic Language as Scaling Technology. When human groups exceeded Dunbar's number (approximately 150 individuals), gossip-based reputation tracking and dyadic partner selection became computationally intractable. The evolution of prescriptive language, specifically deontic modality (must, may, should), allowed proto-norms to abstract, generalize, and scale. As I argued in "Before the Command Was Spoken," deontic language co-evolved with third-party punishment: without verbal articulation of rules, third-party enforcement is impossible, because the enforcer cannot invoke a shared norm that exists only as distributed dyadic knowledge. With language, anyone can cite the rule and thereby legitimate their intervention. This stage explains a transition that Egeland et al. recognize but do not develop: how the individual-level coalitional value exchange described by the CIH scaled into institutional form. An important methodological caveat is necessary here. The transition from Stage 2 (pre-linguistic proto-norms) to Stage 3 (deontic modality) is the least directly evidenced step in the genealogy. Proto-norms leave no archaeological trace of their linguistic formulation; what we observe are indirect indicators (cross-linguistic universality of deontic categories, ontogenetic priority of deontic over epistemic modality, archaeological signatures of behavioral complexity). The inference is consilient but not direct. Future archaeological and paleolinguistic research may refine, or challenge, the specific temporal and causal ordering proposed here.

Stage 4: Formal Law as Institutionalized Coalitional Value Exchange. Once prescriptive language enabled norm articulation, the cognitive-computational services provisioned by intelligent individuals could be codified, transmitted, and enforced by specialists. The jurisconsult, the mediator, the judge: all are institutional elaborations of the role described by the CIH. Legal codes are repositories of accumulated coordination solutions. Courts are institutional mechanisms for the ongoing provision of cognitive-computational services (dispute resolution, norm clarification, collective action coordination). Legal education is phenotypic engineering: the systematic modification of cognitive architecture to produce effective replicators of the dominant legal memeplex. And the prestige hierarchy within the legal profession mirrors the coalitional value system that Egeland et al. describe: status accrues to individuals perceived as provisioning high-quality cognitive-computational services (brilliant advocates, influential scholars, precedent-setting judges), while those perceived as mere formal replicators occupy lower positions.

This four-stage account integrates the CIH's evolutionary psychology with EPT's institutional analysis. The CIH proposes *why* humans evolved the cognitive capacities necessary for law (Stage 1). EPT proposes *how* those capacities were phenotypically expressed before language (Stage 2), *how* they scaled beyond small groups (Stage 3), and *how* they became institutionalized into formal legal systems (Stage 4).

4. The Scaling Problem and the Honesty Paradox

The integration raises a specific analytical challenge that neither the CIH nor EPT had previously addressed: the honesty paradox of institutionalized law.

Egeland et al. insist that the behavioral manifestations of intelligence function as *honest* signals of coalitional value. This honesty is a necessary condition for the coalitional value system to function; if signals could be cheaply faked, cheaters would exploit the system, driving cooperators to extinction (Price, 2006; Trivers, 1971). The CIH's second prediction, that people can gauge the intelligence levels of others with significant accuracy, supports the claim that the signal-receiver system maintains sufficient reliability for the exchange to persist.

In small-scale societies, honesty is maintained by direct observation. If a mediator consistently produces poor outcomes, coalition members observe this directly and withdraw deference. The feedback loop is tight: signal (cognitive performance), assessment (coalitional value gauge), and consequence (status adjustment) operate within the same social environment and temporal frame.

Formal legal institutions, however, introduce three sources of signal degradation that can undermine honesty.

First, *assessment displacement*: legal competence is evaluated not by direct observation of coordination outcomes but by proxy measures (exam performance, publication record, judicial appointments) that can diverge from genuine problem-solving capacity. A legal scholar whose work is heavily cited but produces no measurable improvement in institutional coordination still accumulates prestige. The coalitional value gauge, evolved for face-to-face assessment, operates on proxies that the gauge was not designed to evaluate.

Second, *temporal decoupling*: in small-scale societies, the cognitive-computational service and the prestige reward occur within the same interaction or closely thereafter. In formal legal systems, the reward (appointment, promotion, citation) may follow the service by years or decades, and may persist long after the individual has ceased provisioning useful services. Emeritus status, lifetime judicial appointments, and canonical doctrinal authority all represent temporal decoupling between service and reward.

Third, *memeplex capture*: as I analyzed through the concept of Parasitic Spontaneous Order (PSO), the extended phenotypes of dominant legal memeplexes can capture the prestige allocation mechanism. When law school curricula, peer-reviewed journals, and bar examinations all evaluate conformity with the dominant memeplex rather than genuine cognitive contribution to coordination problems, the system rewards fidelity of replication over quality of innovation. The coalitional value system, designed by evolution to reward honest signals of cognitive capacity, is co-opted to reward honest signals of memeplex loyalty.

The Altruism Paradox paper (Lerer, 2025b) provides partial resolution: in domains where repeated interaction among identifiable actors persists (the *lex mercatoria*, arbitration communities, specialized professional networks), reputational mechanisms continue to track genuine cognitive contribution with reasonable fidelity. Merchant communities resist state intervention in their dispute resolution precisely because the coalitional value system operates more effectively in repeat-player environments where signals remain honest. The Jewish diamond merchant network of 47th Street in New York, the Lloyd's of London insurance market, the international commercial arbitration community: all are institutional environments where the evolved coalitional value gauge still functions approximately as designed.

The prediction that follows is specific: legal institutions should exhibit a measurable honesty gradient, with repeat-player environments (specialized arbitration, commercial law, professional disciplinary proceedings) showing tighter coupling between cognitive-computational service quality and prestige allocation than one-shot environments (criminal defense for indigent clients, mass tort litigation, constitutional adjudication). This is testable through comparative analysis of career trajectories, reputational rankings, and outcome quality across legal subdomains.

5. Novel Predictions from the Integrated Framework

The integrated CIH-EPT framework generates three predictions that are novel to the integration and not derivable from either framework alone.

5.1. Prediction 1: Coalitional Value Signatures in Legal Institutions

If law originated as the institutionalization of cognitive-computational services exchanged for prestige within the coalitional value system, then the institutional core of legal systems should preserve structural features of that exchange. Specifically, the most evolutionarily conserved features of legal institutions should be those that facilitate the identification, evaluation, and rewarding of cognitive contribution to coordination problems.

Operationalization: Compare institutional features across 20 legal systems (10 common law, 10 civil law) spanning at least 200 years of documented history. Code for the presence of (a) mechanisms for identifying individual cognitive contributors (named opinions, attributed doctrine, transparent reasoning), (b) prestige allocation tied to coordination outcomes (promotion criteria, citation patterns, reputational surveys), and (c) accountability mechanisms for poor cognitive performance (reversal rates, disciplinary proceedings, reputational sanctions). Predict that features (a)-(c) will be present in at least 85% of systems, across legal traditions, and that their absence will correlate with institutional dysfunction (measured by rule-of-law indices, litigation duration, and user satisfaction surveys).

Falsification criterion: Fewer than 60% of systems exhibit all three features, or no correlation between feature presence and institutional performance.

5.2. Prediction 2: Prestige Tracks Cognitive Service Quality, Not Credentials Alone

If prestige in legal professions reflects the coalitional value system described by Egeland et al., then status allocation should correlate with the capacity to provision cognitive-computational services (innovative problem-solving, effective coordination, novel doctrinal contributions) rather than with formal credentials alone (degree prestige, years of experience, institutional affiliation).

Operationalization: In a sample of 200 legal professionals across 4 jurisdictions, measure (a) prestige (peer ratings, citation counts, media visibility), (b) formal credentials (law school ranking, years since admission, firm/court prestige), and (c) cognitive-computational service quality (client outcome quality-adjusted, novel doctrinal contributions, coordination innovations adopted by others). Predict that (c) will explain incremental variance in (a) beyond (b), with partial correlation $r > 0.30$ after controlling for credentials.

Falsification criterion: Cognitive-computational service quality does not predict prestige after controlling for formal credentials (partial $r < 0.15$, $p > 0.10$), suggesting the coalitional value system has been fully captured by credentialism.

5.3. Prediction 3: Legal Transplant Failure Correlates with Coalitional Value Mismatch

If legal transplant success depends on compatibility between the coalitional value systems of source and recipient societies (as both the CIH and EPT predict), then transplant failures should correlate specifically with mismatches in how cognitive-computational services are evaluated and rewarded, rather than with formal institutional differences alone.

Operationalization: In a dataset of 40 legal transplant episodes (20 successful, 20 failed), code source and recipient societies for (a) prestige allocation mechanisms (dominance-based vs. prestige-based; individualistic vs. collective; merit-based vs. ascription-based), (b) cognitive-computational service valuation (which services receive highest prestige: adjudication, mediation, legislation, scholarly commentary), and (c) accountability structures (direct observation vs. proxy evaluation; immediate vs. delayed feedback). Predict that coalitional value system mismatch (composite of a-c) will predict transplant failure (odds ratio > 3.0, $p < 0.01$) with greater accuracy than formal institutional similarity measures (legal family classification, constitutional structure, procedural similarity).

Falsification criterion: Coalitional value system mismatch does not predict transplant failure (OR < 1.5, $p > 0.10$), or formal institutional similarity is a stronger predictor.

Data availability note. The dataset of 40 legal transplant episodes required for testing Prediction 3 does not yet exist in the form specified above. Candidate cases can be drawn from existing comparative law databases (Berkowitz, Pistor, and Richard, 2003; Siems, 2014) and the legal origins literature (La Porta et al., 1998, 2008), but the coalitional value system coding (dimensions a-c) requires original fieldwork combining ethnographic, institutional, and survey methods. This prediction is therefore prospective: it specifies what a future empirical test should measure and what results would falsify the hypothesis. Preliminary case identification and coding protocols are available from the author upon request.

Boundary conditions for the integrated framework. The CIH and EPT do not have equal explanatory leverage across all institutional contexts. Three conditions determine which framework dominates. First, when interaction frequency is high and group size is small (repeat-player environments, specialized professional communities, arbitration networks), the CIH's honest signaling mechanism operates with minimal degradation; here the coalitional value gauge tracks genuine cognitive contribution, and the primary predictions of the CIH apply directly. Second, when interaction frequency is low and group size is large (mass litigation, regulatory compliance across anonymous populations, constitutional adjudication affecting millions), signal degradation becomes severe; here EPT's emphasis on memetic competition, phenotypic engineering, and lock-in dynamics provides greater analytical traction. Third, in transitional contexts where institutions are being created or reformed (legal transplants, post-crisis institutional design, emerging regulatory domains), both frameworks contribute: the CIH predicts which institutional designs will attract coalitional investment (those that reward cognitive contribution), while EPT predicts which designs will persist (those whose memes build self-reinforcing phenotypes). These boundary conditions are themselves testable: the honesty gradient predicted in Section 4 provides an empirical measure of where each framework dominates.

6. Discussion and Limitations

Several limitations must be acknowledged. First, the CIH itself remains a hypothesis with testable predictions, some of which have not yet been tested. The integration I propose inherits whatever empirical uncertainty attaches to the CIH. This inheritance is particularly consequential for one line of argument: Section 2.4, where I connect legal citations to the CIH's Prediction 3 (adaptive encoding of intelligence information), relies on a prediction that Egeland et al. themselves acknowledge has not been experimentally confirmed. If future research disconfirms adaptive encoding of intelligence information, the specific argument about legal citations as reputational signals would require revision, though the broader argument about law originating in coalitional value exchange would survive on the strength of the CIH's other, better-supported predictions. The EPT framework itself would remain supported by independent evidence, including the 93.3% functional extinction of the Nineteenth Amendment despite formal constitutional validity (Lerer, 2025d) and Argentina's fiscal lock-in (Lerer, 2025c).

Second, the four-stage genealogical account I develop in Section 3 is necessarily reconstructive. We cannot directly observe the transition from coalitional value exchange in small-scale societies to formal legal institutions. The account relies on consilience across multiple evidence streams (primatology, archaeology, ethnography, evolutionary psychology, comparative law) rather than on any single definitive test. This is appropriate for evolutionary reconstruction, but it means that the account is underdetermined by available evidence.

Third, the honesty paradox identified in Section 4 raises a genuine tension. If formalized legal institutions can decouple signal from substance (rewarding memeplex loyalty over genuine cognitive contribution), then the coalitional value system may be a better explanation for the *origins* of law than for its *current* operation. In highly institutionalized settings, EPT's emphasis on memetic competition and phenotypic engineering may provide more explanatory leverage than the CIH's focus on honest signaling. The two frameworks are complementary rather than competing, but the boundary conditions for each require further specification.

Despite these limitations, the integration generates clear empirical commitments. The three predictions in Section 5 are falsifiable, operationalized with specific measurement protocols, and distinguishable from predictions of rival frameworks. If confirmed, they would provide the first integrated evolutionary account linking the cognitive capacities that produced law (CIH), the pre-linguistic normative patterns that preceded law (EPT), the scaling mechanisms that enabled law to govern large populations (deontic language as transmission technology), and the institutional dynamics that sustain or undermine law (memetic competition and phenotypic engineering).

The broader implication is methodological. Legal theory has traditionally operated in isolation from evolutionary science. The integration proposed here suggests that isolation is no longer sustainable. Law is not an autonomous domain requiring only internal analysis; it is a biological and cultural phenomenon whose origins, persistence, and change are governed by the same evolutionary dynamics that govern all complex adaptive systems. Understanding law requires the same consilience across disciplines that Egeland et al. demonstrate for understanding intelligence: psychology, anthropology, cognitive science, comparative law, institutional economics, and evolutionary biology, working together rather than in parallel.

7. Conclusion

Law did not appear when a sovereign first issued a command. If the argument of this paper is correct, it appeared when an intelligent individual first resolved a dispute within a coalition and received deference in return. The CIH of Egeland, Kennair, and Kleppestø (2026) proposes the evolutionary mechanism that produced such individuals: intelligence evolved as an honest signal of coalitional value, exchanged for prestige in a system of reciprocal cooperation. Extended Phenotype Theory proposes what happened next: the normative solutions those individuals provisioned were encoded in behavioral patterns, amplified by prescriptive language, and institutionalized into formal legal systems that persist as extended phenotypes of competing normative memes.

The integration of these frameworks produces a genealogical account of legal origins that aspires to be both more complete and more testable than either achieves alone. The CIH proposes why natural selection favored the specific cognitive capacities deployed in legal reasoning. EPT proposes how those capacities were phenotypically expressed, memetically transmitted, and institutionally elaborated across evolutionary time. Together, they suggest that law is not a human invention imposed upon nature, but a natural phenomenon, the extended phenotype of memes that have been competing for replication since before our ancestors could speak.

The implications extend beyond theory. If legal institutions are evolutionarily grounded in the coalitional value exchange, then institutional design should attend to the conditions under which that exchange functions effectively: honest signaling, accountability, direct observation, and repeat interaction. Institutions that degrade these conditions, through credentialism, temporal decoupling, or memeplex capture, may progressively lose their capacity to provision the cognitive-computational services that justify their existence. The diagnosis is evolutionary; the prescription is institutional: restore the conditions under which the coalitional value system that created law can continue to sustain it.

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